

COMPUTER NETWORK Model Pratical

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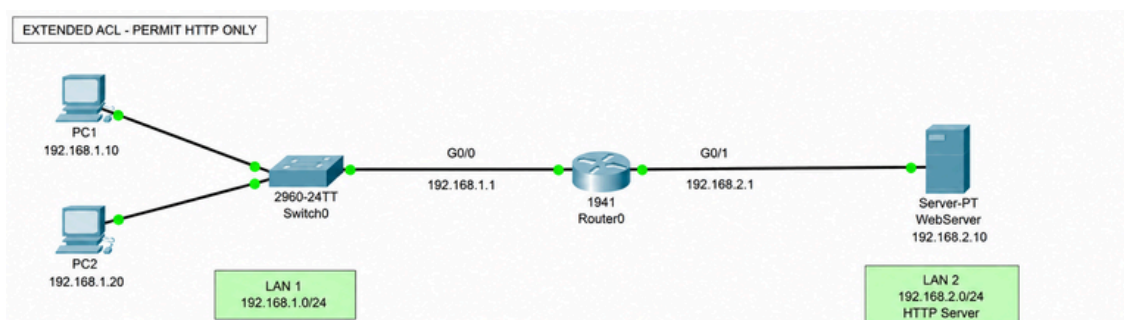
Reg.No:1972260016

SET 6

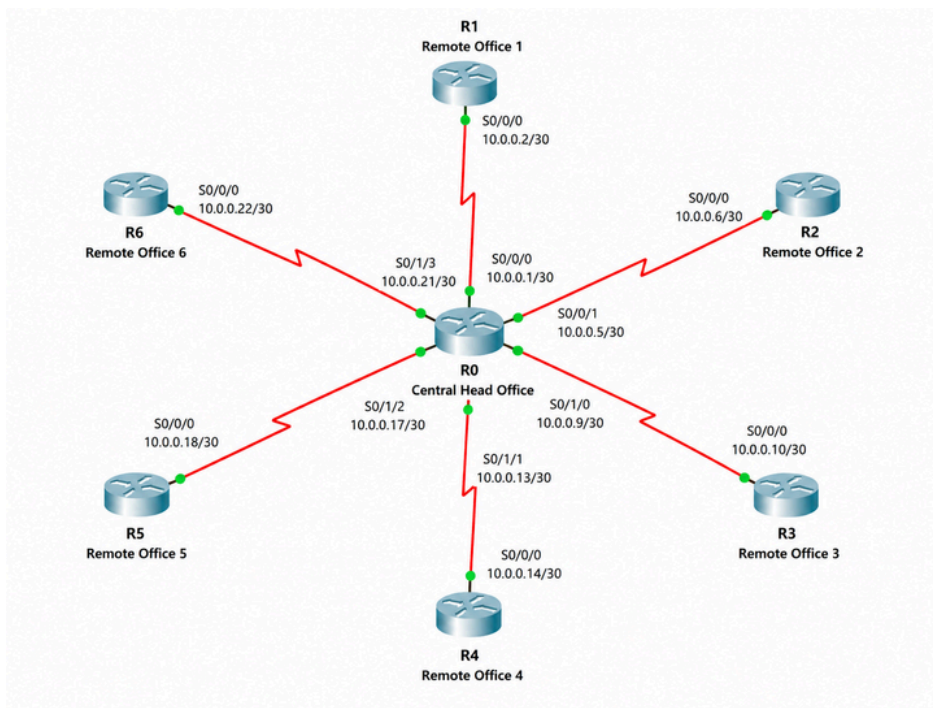
1. Build a network in Packet Tracer using an Extended Access Control List (ACL) to permit only HTTP traffic from a specific subnet to a web server, while blocking all other traffic.
2. Assume that you are the network administrator required to connect 6 remote offices to a central head office using a suitable WAN topology. Design and justify your topology choice using Packet Tracer.
3. Using Wireshark, Capture and analyze the Packets that use NTP (Network Time Protocol) for different header fields and payload of the concerned packet.
4. Using C program, Implement and demonstrate the Cyclic Redundancy Check (CRC) for a given data frame using a generator polynomial of your choice.

MODEL PRACTICAL EXAM

QUESTION 1:



QUESTION - 2



QUESTION 3:

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ntp

No.	Time	Source	Destination	Protocol	Length	Info
102	23.843570	192.168.1.105	192.168.1.1	NTP	90	NTP Version 4, client
103	23.845325	192.168.1.1	192.168.1.105	NTP	90	NTP Version 4, server
104	83.845438	192.168.1.105	192.168.1.1	NTP	90	NTP Version 4, client
105	83.847233	192.168.1.1	192.168.1.105	NTP	90	NTP Version 4, server
106	143.847450	192.168.1.105	192.168.1.1	NTP	90	NTP Version 4, client
107	143.849198	192.168.1.1	192.168.1.105	NTP	90	NTP Version 4, server
108	203.849356	192.168.1.105	192.168.1.1	NTP	90	NTP Version 4, client
109	203.851134	192.168.1.1	192.168.1.105	NTP	90	NTP Version 4, server

> Frame 103: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface \Device\NPF_{CAF8D08C7-8C84-4B6F-A1D5-7E6F8E1C1F3A}, id 0

> Ethernet II, Src: Cisco_7b:3c:7a (24:db:ac:7b:3c:7a), Dst: IntelCor_12:34:56 (3c:52:82:12:34:56)

> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.105

> User Datagram Protocol, Src Port: 123, Dst Port: 62321

Network Time Protocol (NTP Version 4, server)

Leap Indicator: no warning (0)

Version: NTP Version 4 (4)

Mode: server (4)

Stratum: 2

Polling Interval: 6 (64 s)

Peer Clock Precision: -20 (0.000001 s)

Root Delay: 0.00025298 s

Root Dispersion: 0.00010812 s

Reference ID: C0A80101 (192.168.1.1)

Reference Timestamp: Jun 02, 2025 03:19:40.000000000 IST

Originate Timestamp: Jun 02, 2025 03:20:43.843570000 IST

Receive Timestamp: Jun 02, 2025 03:20:43.845123000 IST

Transmit Timestamp: Jun 02, 2025 03:20:43.845325000 IST

0000 3c 52 82 12 34 56 24 db ac 7b 3c 7a 08 00 45 00 <R..4V\$.-(<z. E.

0010 00 4c 1c 46 40 00 40 11 e2 b3 c0 a8 01 01 c0 a8 ..L.F@.@.....

0020 01 69 00 7b f3 69 e5 5a 00 38 3c d8 23 00 04 24 .i.{.i.Z.8<-.#..\$

0030 02 06 ec 14 00 00 00 00 00 00 00 c0 a8 01 01C.....

0040 ec 0d 43 20 e9 06 02 03 e9 06 02 03 e9 06 02 03 ..C.....

0050 00 00 00 00

wireshark_Wi-Fi_2050602_032015.pcapng

Packets: 1242 • Displayed: 8 (0.6%) • Dropped: 0 (0.0%)

Profile: Default

QUESTION 4:

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 void xorOperation(char *data, char *generator, int pos, int genLen) { 5 int i; 6 7 for (i = 0; i < genLen; i++) { 8 data[pos + i] = 9 (data[pos + i] == generator[i]) ? '0' : '1'; 10 } 11 } 12 13 int main() {</pre>		<pre>Enter data bits: 1101011011 Enter generator polynomial: 10011 Data after appending zeros: 11010110110000 CRC remainder: 1110 Transmitted frame: 11010110111110 === Code Execution Successful ===</pre>